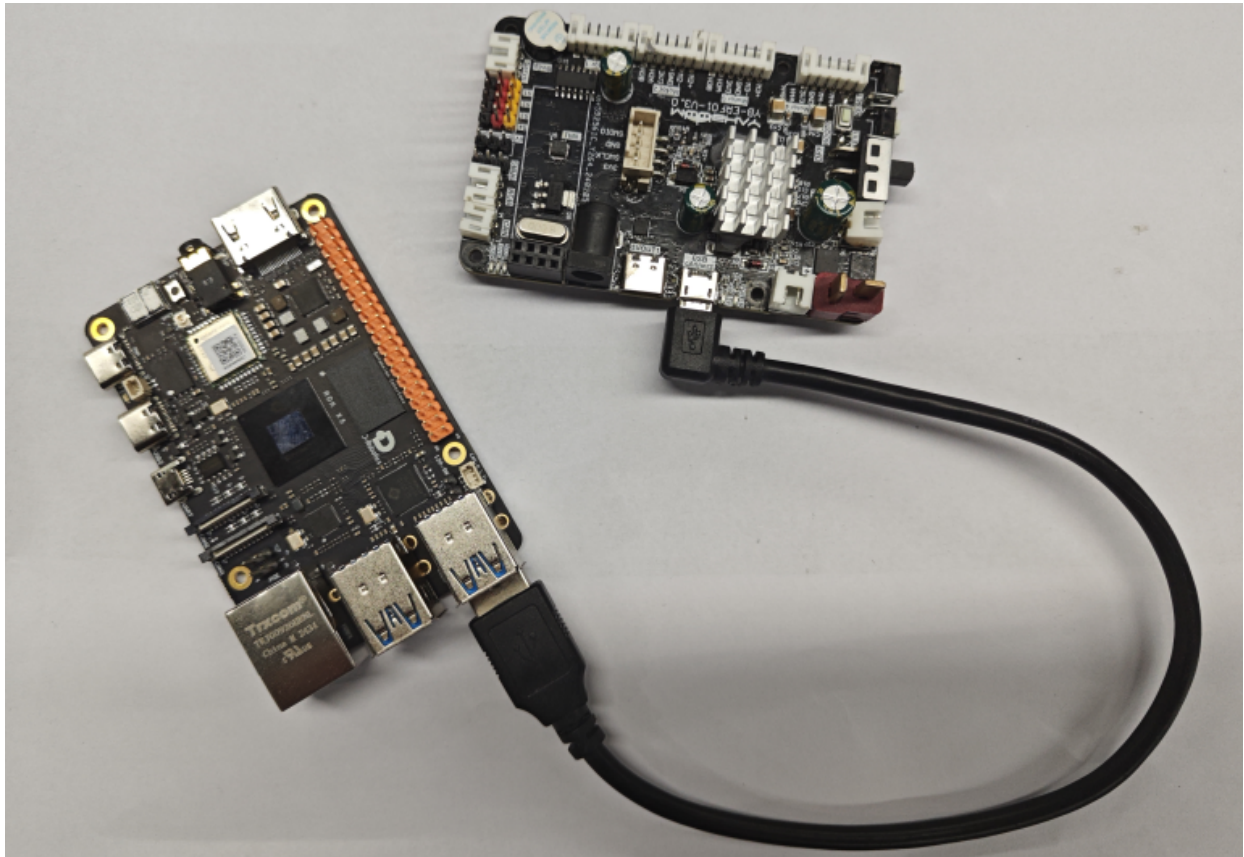


Drive OLED

The purpose of this tutorial is to display the IP address of RDK X5 through serial communication on the ROS expansion board. Before running, please confirm that the firmware of the expansion board has been updated.

1.Preparation before use

Need to use ROS expansion board and X5 and USB data cables. The following diagram is the wiring diagram,

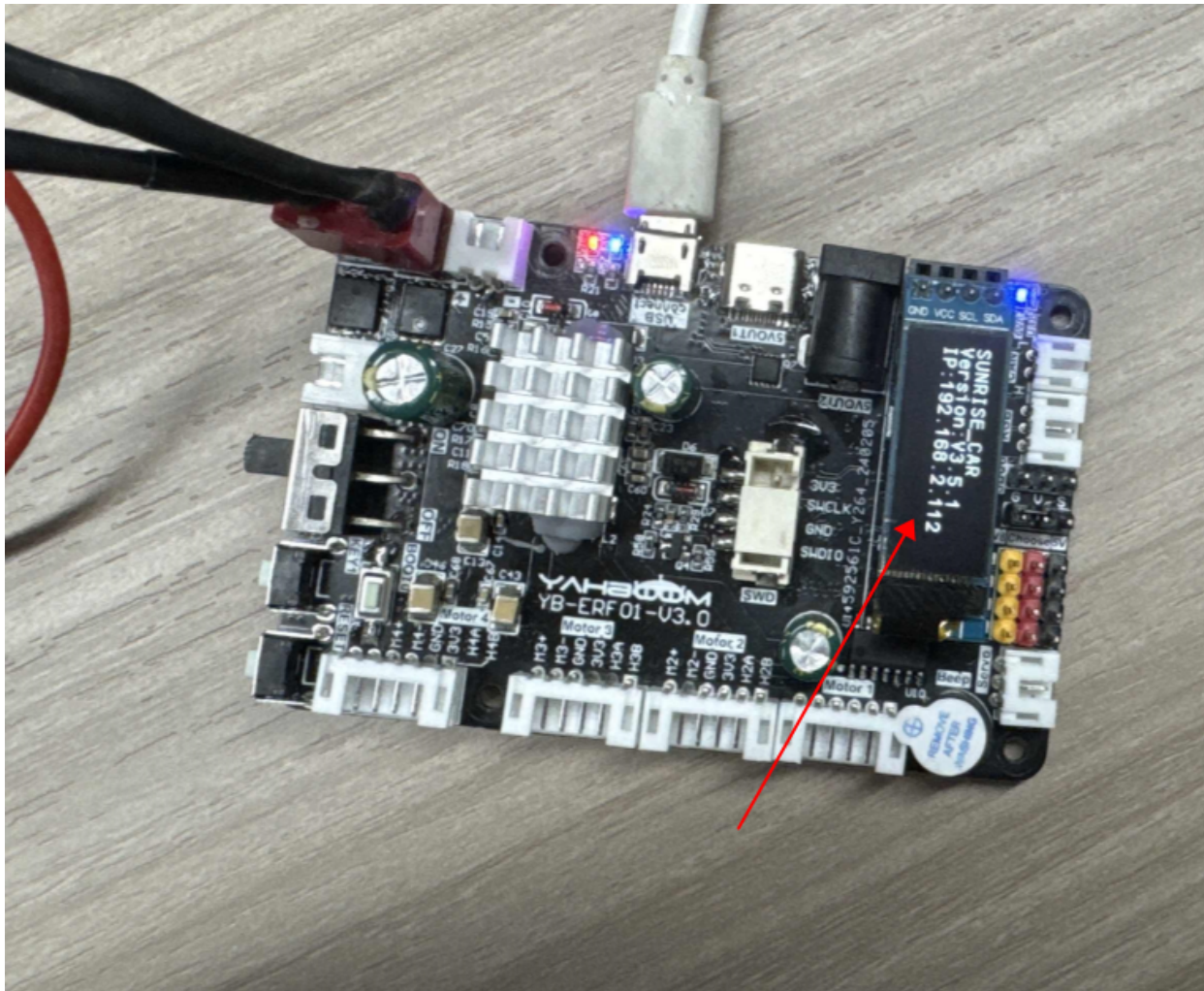


When we plug in the battery, we can see the OLED printing firmware version number and other information.


```
python3 oled.py
```

```
sunrise@ubuntu:~$ python3 oled.py  
sunrise@ubuntu:~$
```

At this point, the OLED screen displays information such as the IP address of the X3, firmware version of the ROS control board, and current vehicle model.



It is known that ip: 192.168.2.112 is the current address of x5 dispatch.

3.Code Interpretation

Import serial port module, network programming module, etc

```
import socket, struct, fcntl  
from time import sleep  
import serial
```

Get the current host IP address

```
hostname = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    return
socket.inet_ntoa(fcntl.ioctl(hostname.fileno(), 0x8915, struct.pack('256s', bytes(ifname[:15], 'utf-8')))[20:24]))
```

Decompose IP address data

```
def split_ip(ip):
    parts = ip.split('.')
    return parts
```

Transfer the current address via serial port

```
cmd = [0xFF, 0xFC, 0x07, 0x71, part_L, part1_L, part2_L, part3_L]
checksum = sum(cmd, 5) & 0xFF
cmd.append(checksum)
ser.write(cmd)
```